Universität Hamburg
DER FORSCHUNG | DER LEHRE | DER BILDUNG

Wirtschafts – und Sozialwissenschaften

MSc in Economics

Prof. Melanie Krause, Ph.D.

URBAN ECONOMICS – Master's Seminar

REPLICATION PAPER WITH EXTENSION

"IS UBER A SUBSTITUTE OR COMPLEMENT FOR PUBLIC TRANSIT?"

Jonathan D. Hall, Craig Palsson, Joseph Price

Submitted by

Lisa Fadelli

bay7535@studium.uni-hamburg.de

ID: 7374198

14th February 2022

WiSe 2021/2022

TABLE OF CONTENTS

1. ABSTRACT 3
2. INTRODUCTION 3
3. LITERATURE REVIEW 4
4. REPLICATION OF THE PAPER 5
4.1. DATA AND METHODS 5
4.2 MAIN ANALYSIS 7
4.3 COMMENTS ON REPLICATION 10
5. EXTENSION 10
5.1 DATA AND METHOD 10
5.2 ECONOMETRIC ANALYSIS 12
5.3 RESULTS AND CRITICISMS 13
6. CONCLUSION 13
7. BIBLIOGRAPHY 14

1. ABSTRACT

The study conducted by Hall and et. has shown that Uber is a complement for the average transit agency, increasing ridership after two years from the entry date into the Metropolitan Statistical Area in the United States, and the effect grew over time.

With the outbreak of the Covid-19 pandemic, a reduction in traffic was observed worldwide (Statista, 2021), and the extension part of this paper tries to investigate the effect of new weekly Covid-19 cases on Uber usage in several MSAs across the U.S.., using the Google Trend search intensity as a proxy. The concluding results show a tiny decrease in the Google Trend index as the cases increased.

2. INTRODUCTION

Uber is a ride-hailing start-up company born in March 2009 in San Francisco in the United States that represents one of the main taxi industry competitors nowadays, together with Lyft (Cramer & Krueger, 2016). Where Uber operates, the client can request a trip through the Uber app which is automatically sent to the nearest Uber driver. The app also provides all information related to the driver-partner, including the first name, type of vehicle and number of the license plate (Uber, 2021).

But what is the effect of Uber on public transit? This relevant policy question has been facing by most of the biggest cities worldwide (Hall, et al., 2018).

Can Uber be considered as a complement for public transport or is it more a substitute?

In the paper "Is Uber a substitute or complement for public transit?", the authors underlined that there is a clear relationship between Uber and public transit use: it has been calculated that in several American cities, 25%-40% of Uber pick-ups and drop-offs are near public transit stations (Smith, 2015) and the interaction between Uber and Public transportation system affects congestion and pollution (Hall, et al., 2018).

As the authors pointed out, Uber could affect public transit through 2 mechanisms:

as an alternative way of travel: in this case, riders will substitute public transit for Uber for convenience if the costs are lower than taking taxi-like service; or to overcome the problem caused by fixed schedules. In the latter case, Uber can fill in the holes in public transit coverage and the combination makes it possible to complete all desired trips without owning a car, providing more flexibility.

3. LITERATURE REVIEW

This paper by Hall J.D., Palsson C. and Price J. is comprised in the topic on Transportation. This overall topic tries to answer how the transportation infrastructure shapes the positive and negative aspects of city life.

The first paper from the collection, written by Gonzalez-Navarro and Turner in 2018, analysis to what extent highways caused suburbanization in the US between 1950 and 1990.

The authors used several regressions to detect the causal relationship for this phenomenon: OLS; Long difference, panel analysis and IV using planned portion of the interstate highway system as a source of exogenous variation. This approach ultimately indicated that one new highway passing through a central city reduces its population by about 18% (Gonzalez-Navarro & Turner, 2018).

In this paper, social issues like crime rate and gentrification were discussed and the accuracy of the results was supported by the empirical strategy used, which exploited the fact that an observable portion of the national road system was designed to connect remote locations.

In the American society in certain parts of the US (mainly East-Coast) it is common to commute to work using trains (Burrows, et al., 2021), so further researchers might ask if also the railway system caused suburbanization.

The other paper, “Did highways caused suburbanization?” by Baum-Snow published in 2007, discussed the relationship between subways and urban growth on a worldwide scale, analysing the effect over more than 630 cities. According to the authors, large cities are more likely to have subways; nevertheless, subways have an insignificant effect on urban population growth (Baum-Snow, 2007). This conclusion was drawn by a panel data regression, using the lights at night data to calculate the centralization. This research also evidences that subways cause cities to decentralized, although the effect is smaller than previously documented effects of highways on decentralization.

In large cities where Uber or any other ride-hailing company is present, the policymakers are struggling to assess the impact of these companies in terms of mobility and sustainability. That depends, also, on whether they are serving as a substitute or a complement to public transit (Singapore - MIT Alliance for Research and Technology, 2020). To investigate more the relationship between ride-sharing and public transit, a recent study conducted by the SMART Future Urban Mobility interdisciplinary research group, together with the JTL Urban Mobility Lab at MIT, found that in Chengdu China, a major urban centre with more than 16 million

inhabitants, one-third of DiDi[1] trips potentially substitutes public transit. The researchers have also found that this substitution effect is more significant in areas served by subways service, whereas in peripheral and suburban areas the ride-hailing company acts more as a complement (Singapore - MIT Alliance for Research and Technology, 2020).

4. REPLICATION OF THE PAPER

4.1. DATA AND METHODS

To estimate the effect of Uber on public transit ridership, the authors used a difference-in-difference approach.

According to this estimation method, the aim is to compare how transit ridership changes in the cities when Uber enters relative to changes in cities where Uber has not entered. Uber Company offers a wide variety of services, so the authors focused their attention on the entry of UberX, which accounts for the majority of their ridership.

For what concerns the data collection, to estimate the effect of Uber on public transit the unit of observation is the travel agency: the sample contains 196 Metropolitan Statistical Areas where Uber was present, and each MSA includes on average 2.21 transit agencies. The data on the transit ridership was taken from the National Transit Database: this database contains the number of times a rider uses a transit vehicle, reporting the monthly ridership for all transit agencies separated by mode (bus, train, etc.). The time frame of this research covered 11 years, from 2004 to 2015.

	Mean	Median	Std. dev.
Panel A: Transit agency summary statistics			
Average bus fare	1.31	0.75	4.45
Average rail fare	4.94	2.00	11.0
Average fare	1.77	0.91	4.13
Maximum bus vehicles operating	108.1	30	283.8
Maximum rail vehicles operating	274.5	51	725.3
Maximum vehicles operating	185.2	48	590.1
Bus vehicle revenue hours (1,000)	27.8	6.90	79.1
Rail vehicle revenue hours (1,000)	76.5	11.9	231.2
Vehicle revenue hours (1,000)	40.4	8.70	158.9
Bus vehicle revenue miles (1,000)	345.6	100.4	813.7
Rail vehicle revenue miles (1,000)	1672.0	231.1	4556.6
Vehicle revenue miles (1,000)	616.5	135.2	2326.7
Bus ridership (100,000)	9.53	1.32	41.3
Rail ridership (100,000)	63.5	7.90	263.8
Total ridership (100,000)	15.6	1.21	124.7
Observations	76,213		
Panel B: MSA summary statistics			
Population (100,000)	8.74	3.45	17.9
Employment (100,000)	4.12	1.57	8.39
Gas price	2.94	2.91	0.65
Google search intensity for "Uber"	2.08	0.75	3.79
Uber in MSA	0.075	0	0.26
Observations	40,012		

Table 1: Summary Statistics

[1] DiDi is a transportation network company which provides "on-demand services that use mobile devices and applications to connect drivers with passengers" (Kong, et al., 2020)

The estimates are based on tis regression:

$$Y_{i,t} = \beta D_{c(i),t} + \gamma_i + \delta_t + \theta_{i,m(t)} + \zeta_{c(i)} * t + X'_{i,t}\eta + \epsilon_{i,t}$$

where $Y_{i,t}$ represents the log transit ridership on transit agency i in year-month t; $D_{c(i),t}$ is a dummy variable which stands for 1 if UberX is active in the $MSA_{c(i)}$ in year-month t and 0 otherwise; γ_i is a transit agency specific fixed effects; δ_t is the year-month specific fixed effect; $\theta_{i,m(t)}$ is the transit agency-calendar month fixed effect included in order to address seasonality in transit ridership; $\zeta_{c(i)}$ is the MSA specific time trend; and $X'_{i,t}$ is a vector of controls for transit agency i.

To detect the Uber variable, the authors used two sources of variation: when Uber entered each market and the variation of intensity of Uber penetration, estimating two versions of the same empirical model. For the former source of variation, Uber's Entry and Exit dates were gathered from newspaper articles as well as press releases, blog posts, and social media posts, while for the latter, they used the relative number of Google searches for the word "Uber" in each MSAs. The search intensity was then normalized by the search index in San Francisco on a specific week (29th of January 2015) and converted into standard deviation units.

As can also be seen in Figure 1, this measure is strongly correlated with the number of Uber active drivers per 10,000 residents in each of the reported markets.

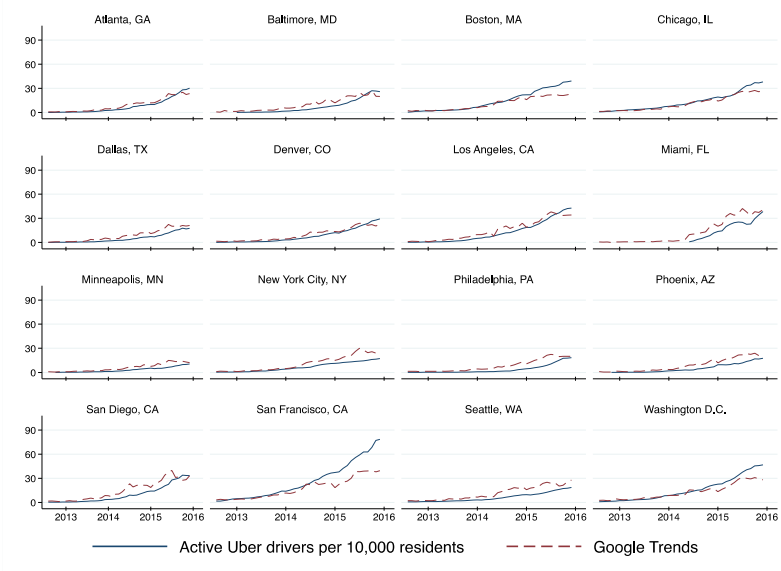

Figure 1: Relative frequency of Google searches for "Uber" compared to number of active Uber drivers per 10,000 residents.

The control variables enclosed in the dataset are MSA population, income, age and education in regressions predicting Uber's entry decision, while in regression estimating the effect of Uber, monthly MSA total employment and unemployment rate, annual MSA population and monthly regional gas prices were used.

Since the controls for transit agency might change endogenously in response to Uber's entry, the authors add the robustness condition, and the standard errors are clustered at the MSA level. For Uber's variation variable, the choice between the two (Uber Entry&Exit date or Uber penetration) is a trade-off between interpretation and precision. Despite this, the authors claim that together they provide a good insight.

In the Difference-in -Differences econometric design, there is the key identifying assumption: parallel trends assumption, according to which the trend would be the same in treated and untreated cities before the event occurred (Wooldridge, 2018). This assumption has been checked by studying the event study plot for each regression.

4.2 MAIN ANALYSIS

When the board of directors was planning the business idea, Uber's purpose was to reach most of the nation and the world as soon as possible.

Figure 2 shows that Uber primarily entered markets according to population rank order, from large to small.

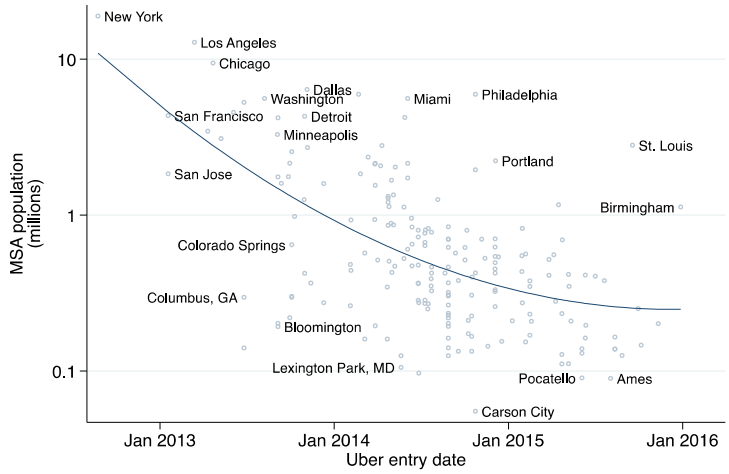

Figure 2: Uber entry date by MSA population

The following table reports the result of the first two regressions conducted by the authors.

In the first column, the coefficients from the linear regression predicting when Uber enters an MSA are reported. Note that the independent variables are measured in standard deviation units to facilitate the comparison of the magnitudes of the coefficients.

From the data, it can be shown that population was the strongest predictor of when Uber enters an MSA; furthermore, Uber was more interested in entering in already big cities first, rather than growing cities, as can be deduced by looking at the coefficient for the Trend in

log(population). The second column reports the coefficients of the linear regression which predicts whether Uber has entered an MSA. The population is still the strongest predictor.

	Date UberX entry	Did UberX enter
	(1)	(2)
Log(population)	-103.2***	0.256***
	(4.341)	(0.00707)
Percent with bachelor's degree	-41.77***	0.180***
	(4.880)	(0.00940)
Median age	30.90***	-0.0518***
	(5.718)	(0.00883)
Median income	-11.40**	-0.0288***
	(4.699)	(0.0102)
Excess unemployment	-41.73***	0.0336***
	(4.424)	(0.00734)
Percent work trips transit	-9.956*	-0.0792***
	(5.207)	(0.00913)
Capital expenditures on public transit	-4.868	-0.00152
	(5.233)	(0.00792)
Dist from Uber HQ	11.99***	0.00823
	(4.427)	(0.00717)
Trend in log(population)	11.46	0.0214*
	(8.489)	(0.0112)
Trend in median income	1.641	-0.0298
	(14.17)	(0.0195)
Observations	197	386
Adjusted R-squared	0.383	0.394

Table 2: Linear regressions predicting when and whether Uber enters an MSA

Table 2 returns the effect of Uber on overall transit ridership. The dependent variable is measured in logs and represents the percent increase in public transit ridership due to the arrival of Uber or the change in the Uber penetration variable.

	(1)	(2)	(3)		(4)	(5)	(6)
UberX	0.00263	-0.0591**	0.0598**	Google Trends	0.0138***	-0.00483	0.0328***
	(0.0143)	(0.0295)	(0.0236)		(0.00515)	(0.00526)	(0.00652)
Above median population x UberX		0.0666**		Above median population x Google Trends		0.0228***	
		(0.0294)				(0.00716)	
Above median ridership x UberX			-0.0811***	Above median ridership x Google Trends			-0.0281***
			(0.0292)				(0.00977)
Agency fixed effects	Yes	Yes	Yes	Agency fixed effects	Yes	Yes	Yes
Month-year fixed effects	Yes	Yes	Yes	Month-year fixed effects	Yes	Yes	Yes
Linear time trend	Yes	Yes	Yes	Linear time trend	Yes	Yes	Yes
Observations	71386	71386	71386	Observations	58015	58015	58015
Clusters	309	309	309	Clusters	227	227	227

Table 3: Effect of Uber on log transit ridership

When Uber enters the MSAs, transit ridership increased by about 0.26% although it is not statistically significant. However, when Uber became more commonly used in MSA, there was an increase in public transit use by almost 1.4%, resulting in the first specification using Uber penetration variable as regressor shows.

The key common trend assumption is fulfilled, as can be deduced also from the event study plot: Figure 3 plots the pre-event and post-event coefficients, given the control variables: since the pre-event coefficients almost stand on the zero line, the common trend assumption is satisfied. The plot shows that transit ridership gradually increases after Uber enters the MSAs.

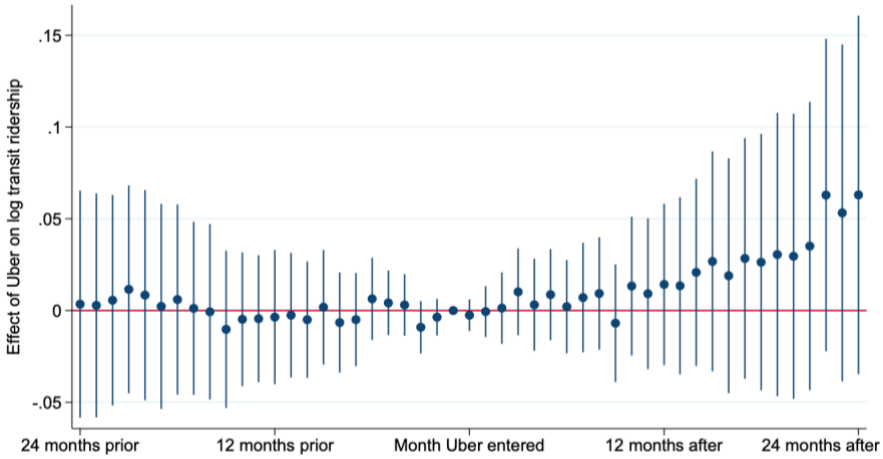

Figure 3: Event study plot - Effect of Uber on log transit ridership before and after entry

The authors then extended the analysis examining how the effect of Uber differed based on the population of MSA and on the number of riders that were using public transit before Uber had arrived. They fixed the median population at 280thousand inhabitants and the median monthly ridership at 82thousand passengers. Then, based on whether the observation was above or below the sample medians, they included each of those binary variables as an interaction term with the Uber measure. The estimations show that in smaller cities, the arrival of Uber has decreased public transit ridership by 5.9%, while for larger cities, it has increased by 0.8% (Table 3 (2)). For transit agencies that had below median public transit ridership, the arrival of Uber increased public transit use by 6%; in contrast, for those agencies above the median, it decreased by 2,1% (Table 3 (3)). All the results mentioned above are statistically significant, at least at the 10% level. The main conclusion is that Uber "most strongly complements small transit agencies in large cities" (Hall, et al., 2018).

Two other hypotheses been tested: how Uber entry affected Uber penetration, using Google Trend as a proxy, and what the effect of Uber is on bus and train ridership. The authors found that Uber entry significantly had a positive effect on the Google Trend search intensity of the word "Uber" (Uber penetration expected to change by 0.470) and consequently increased with the number of transit agencies serving the Metropolitan Statistical Area. The effect of Uber entry on bus ridership are similar to the case on the overall ridership, however the effect on rail

ridership is different. The railway system helps more the larger agencies in large cities, relative to smaller agencies.

4.3 COMMENTS ON REPLICATION

The data for the replication and the codes are available in an open-access database[2].
The results from the replication are almost equal: there are very tiny changes in the standard errors mainly because I decreased the bootstrap repetitions from 1000 to 200.
I replicated the tables 1, 2, 3, 5, 6 of the main analysis, but the eight robustness tests contained in the appendix of the paper.

5. EXTENSION

For the extension part of the paper, I want to develop the analysis of the use of Uber during the Covid19 pandemic in the U.S. from the beginning of March 2020 and the end of August 2021, including the third wave of infections caused by the Delta variant during summer 2021.
The main question is if Uber still acted as a complement to public transportation or if it rather acted as a substitute given its "social distancing" premium characteristic.
This idea came from two different facts: first, measures for preventing the spread of the infections were issued on a state-by-state basis, whereas Uber issued company-wide measures regarding ridership safety (Uber, 2021), and maybe this could have altered the consumer behavior in favor of Uber, given the need of protection and to avoid crowded areas.
Another example, which I found pretty interesting, is that Uber provided free rides to vaccine sites, giving an incentive to the vaccination plan in the U.S. (Uber, 2021).
So, the regression analysis that I implemented is a panel regression which will consider the effect of the new weekly cases on the Google search intensity of the word "Uber", using the search intensity as proxy for Uber use.

5.1 DATA AND METHOD

To obtain data for the Uber penetration, using the relative Google Trend search intensity index, I used the CSV file available from the replication package *Google_Trend_metro_id.csv*. This file contains all the Google Trends reference codes for each Metropolitan Statistical Area in the

[2] The open-access database with the Replication Code and Data
https://www.openicpsr.org/openicpsr/project/115490/version/V3/view;jsessionid=6FE80BFDEE651673A0FCEA2A46F3EDD4

U.S.. I selected 28 MSAs according to the population ranking as of April 1, 2020 (as enumerated by the 2020 United States Census) (Wikipedia, 2022), then I downloaded the trending results from the Google Trends website using a python API. I pulled out the data using Stata and obtained the input dataset file *GTrendsFinal.dta* with all the weekly search intensity for each MSA from the beginning of March 2020 to the end of August 2021.

The explanatory variable of interest is the sum of the new weekly cases in each MSA and I imported the data for the open-access archive from the New York Times website updated to January 2022. This time-series data has been compiled from state and local governments and health departments, e.g. the Center of Disease Control and Prevention, over the months, starting from the first reported coronavirus case in Washington State on the 21st of January 2020 (The New York Times, 2020-2021).

The CSV file contains for all the U.S. counties data about the number of new cases reported that day, the average number of new cases reported in the last seven days, the average number of new cases reported in the last seven days per 100,000 people, the total number of new death from Covid-19, the average number of new deaths reported in the last seven days and the average number of new deaths reported in the last seven days per 100,000 people. Since I am interested in seeing how the Uber search index changes, given the new weekly cases, I dropped some variables and kept just the new daily cases reported that day data as the variable of interest. I selected only the counties included in the MSAs and I merged each MSA to the correspondent counties, using the information found in the National Transit Database.

Since I had to calculate the new weekly cases, I exported a temporary dta file for each MSA and created in each dataset the new variable *weeknr*: integer number to identify the week starting from the 1st January 1960.

I proceeded with the sum of the new covid19 daily cases in each MSA in that particular week, creating the variable *weeklycases*. I merged the Covid19 cases dataset with the dataset containing the search intensity measured as Google Trend index. I sorted the temporary final dataset by MSA id and date, dropping the variables that I will not use in the regression model. The further step was to include a variable that was able to control how the Covid19 pandemic outbreak and the consequent reduction of displacements. The *UPTTotal* variable counts the Unlinked Passenger Trip: the number of passengers who use public transportation vehicles, while the *VOMSTotal* variable stands for Vehicles Operated in Annual Maximum Service: number of revenue vehicles needed to reach the annual maximum service requirement (Federal Transit Administration, 2022). These two variables have been recorded monthly.

Once I collected and merged all data in one unique final dataset, I was ready to run the regression analysis.

	Mean	Median	Std. dev.	Minimum	Maximum
Google Trend Search Intensity	61.6	60	15.0	21.00	100
New weekly cases	6273.8	3323	8315.5	0.00	90715
Unlinked Passenger Trips	9069487.6	3199341	26256640.3	465584.00	197229374
Vehicles Operated in Annual Maximum Service	2048.1	977	3331.4	89.00	19688
Observations	2212				

Table 4: Summary Statistics

5.2 ECONOMETRIC ANALYSIS

The dataset is composed of data collected in 28 MSAs over 79 weeks, therefore I started informing STATA that I am dealing with panel data. STATA provides additional information that the panel is strongly balanced, therefore we have the same time periods *T* for each observation unit *i* (Wooldridge, 2013).

Before starting the econometric analysis, I studied the distribution of the variables. *weeklycases*, *UPTTotal* and *VOMSTotal* had skewed distribution, so I implemented a log transformation.

I proceeded to specify a linear fixed effects model regressing the search intensity of the word "Uber" on the log number of new weekly Covid19 cases reported that specific week, including *lnUPTTotal* and *lnVOMSTotal* as control variables, to account for the general decline in using public transportations due to extensive lockdowns and working-from-home measures.

$$googletrend_{i,t} = \beta_1 + \beta_2 lnweeklycases_{i,t} + \beta_2 lnUPTTotal_{i,t} + \beta_3 lnVOMSTotal_{i,t} + \alpha_i + v_{i,t}$$

The F-test, which checks for the presence of no unobserved heterogeneity, can be rejected. I then implemented the random-effects model. To confirm the hypothesis of unobserved heterogeneity, I ran the Breusch-Pagan LM test for random effects. I used the Hausman Test to check whether the idiosyncratic errors are correlated or not with the regressors. Rejecting the Hausman Test, the fixed effects correlate with the regressors: the random effects model cannot be used in this context since the assumption of cov(α_I, x_{it}) I checked the assumption of no serial correlation in first-differenced idiosyncratic errors implementing the first-difference estimation method and regressing the first-differenced idiosyncratic error on its first lag: I found a highly significant negative correlation. That implies that the model does not present serial correlation in idiosyncratic errors, and thus the fixed effects model seems to be the most suitable to use. To correct the presence of heteroskedasticity, I added the robustness condition *vce(robust).*

5.3 RESULTS AND CRITICISMS

	(1) Google Trend Search Intensity
Log New weekly cases	-1.363***
	(0.278)
Log Unlinked Passenger Trips	10.89***
	(2.502)
Log Vehicles Operated in Annual Maximum Service	2.858
	(4.611)
Constant	-111.5**
	(46.25)
Observations	2182
Adjusted R-squared	0.0718

Standard errors in parentheses
* (p<0.10), ** (p<0.05), *** (p<0.01)

Table 5: Estimation results

Table 5 presents the results of the estimation. A 1% increase in the number of new weekly cases did not present a relevant difference in the Google Trend search intensity index with a decrease by 0,01363 units ceteris paribus, although the result is highly significant. Among the other variables, the one accounting for number of revenue vehicles needed to reach the annual maximum service requirement is not statistically significant.

The adjusted R-squared is 0.0718, which suggests that not much of the variation in the Google Trend Search Intensity is explained by the independent variables. First, I checked for the normal distribution of the idiosyncratic error; then I analyzed the linearity condition of the variables. This resulted in a non perfectly linear relationship between the dependent variable and the explanatory variables, therefore a non-linear fixed-effect model might be desirable, and this finding may be exploited for further research.

It is worth mentioning that, the result might be biased since I incorporated into the sample just the first 28 most populated MSAs, where the probability of being infected is higher, given the high population density and where inhabitants might not have a private vehicle and to move around, they necessarily have to use public transportation or alternative means of transport.

6. CONCLUSION

According to the results presented in the paper of interest, the complementary effect of the ride-hailing company for small transit agencies suggests that consumers use Uber to overcome the fixed route and scheduling problem of the public transportation system.

The analysis I added as an extension provides preliminary insight regarding the Covid-19 pandemic effect on the utilization of Uber as a transport mode in the biggest MSAs across the United States: even though the magnitude of the change is not outstanding, a decrease in the use of Uber has been identified. By looking at the trend of the variable *UPTOtal* in the final dataset used for the econometric analysis, I noticed how the passenger trips in the public transit diminished, as the infections boosted at the beginning of March, most probably together with the implementation of restricting measures. In the following months, the trend did not present remarkable changes.

It has been observed that in American communities where there was a higher proportion of essential jobs and a more heterogeneous society, the minimal level of transportation demand tent to remain high, thus proving that public transit persisted as an essential service during the pandemic (Liu, et al., 2020). Given this, we may infer that Uber continued to serve the biggest cities during the different pandemic phases, also promoting new incentives to get vaccinated.

7. BIBLIOGRAPHY

Baum-Snow, N., 2007. Did highways cause suburbanization?. *Quarterly Journal of Economics,* 122(2), pp. 775-805.

Burrows, M., Burd, C. & McKenzie, B., 2021. *Commuting by Public Transportation in the United States: 2019,* United States Census Bureau: Americal Community Survey Reports.

Cramer, J. & Krueger, A. B., 2016. *Disruptive change in the taxi business: the case of Uber,* Cambridge MA: National Bureau of Economic Research.

Federal Transit Administration, 2022. *National Transit Database NTD Glossary.* [Online] Available at: https://www.transit.dot.gov/ntd/national-transit-database-ntd-glossary [Accessed 12 February 2022].

Gonzalez-Navarro, M. & Turner, M., 2018. Subways and Urban Growth: Evidence from Earth. *Journal of Urban Economics,* 108(C), pp. 85-106.

Hall, J. D., Palsson, C. & Price, J., 2018. Is Uber a substitute or complement for public transit?. *Journal of Urban Economics* , Volume 188, pp. 36-50.

Kong, H., Zhang, X. & Zhao, J., 2020. How does ridesourcing substitute for public transit? A geospatial perspective in Chengdu, China. *Journal of Transport Geography,* Volume 86.

Liu, L., Miller, H. J. & Scheff, J., 2020. The impacts of COVID-19 pandemic on public transit demand in the United States. *PLoS ONE*, 12(11), https://doi.org/10.1371/journal.pone.0242476.

Singapore - MIT Alliance for Research and Technology, 2020. *Does ride-sharing subsitute for or complement public transit?.* [Online] Available at: https://news.mit.edu/2020/how-does-ridesourcing-substitute-public-transit-network-0727
[Accessed 7 February 2021].

Smith, A., 2015. *Using Open Data and Uber Data to Understand How Ridersharing Complements Public Transportation*, Washington D.C.: s.n.

Statista, 2021. *Year-on-year traffic change during a period of national emergency related to coronavirus in selected countries between February and December 2020.* [Online] Available at: https://www.statista.com/statistics/1106135/change-in-daily-traffic-volume-amid-coronavirus-crisis-key-countries/
[Accessed 14 February 2022].

The New York Times, 2020-2021. *Coronavirus (Covid-19) Data in the United States.* [Online] Available at: https://github.com/nytimes/covid-19-data
[Accessed 7 February 2021].

Uber, 2021. *How does Uber work?.* [Online] Available at: https://help.uber.com/riders/article/how-does-uber-work?nodeId=738d1ff7-5fe0-4383-b34c-4a2480efd71e
[Accessed 7 February 2021].

Uber, 2021. *Safety.* [Online] Available at: https://www.uber.com/us/en/safety/
[Accessed 7 February 2021].

Uber, 2021. *Uber Impact.* [Online] Available at: https://www.uber.com/it/en/impact/rides-for-vaccines/
[Accessed 7 February 2021].

Wikipedia, 2022. *Metropolitan Statistica Area.* [Online] Available at: https://en.wikipedia.org/wiki/Metropolitan_statistical_area
[Accessed 11 February 2022].

Wooldridge, J. M., 2013. *Introductory Econometrics A Modern Approach.* 5th Edition ed. Cengage.

Wooldridge, J. M., 2018. *Introductory Econometrics: a modern approach.* 7th Edition ed. Cengage .